Mathematical Analysis - Week 4

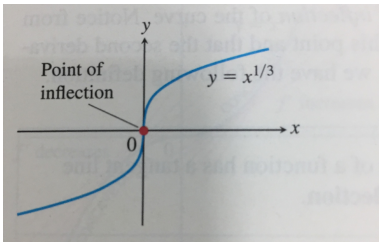
Lecture 13-15, Tutorial 3 and Homework 3

Lecture 13

Inflection and curve sketching

Definition (slides' convention). Inflection point: tangent exists and strict concavity changes across the point. Vertical tangents allowed. $f'' = 0$ or undefined only gives candidates.

f	concavity at 0	tangent	inflection?
$x^{5/3}$	down/up	$y = 0$	yes
$x^{1/3}$	up/down	$x = 0$	yes
x^4	up/up	$y = 0$	no
$x^{2/3}$	down/down	cusp	no



Lecture 13, p.10.

Ex.1.2.46. $f = (x+1)^2/(1+x^2) = 1 + 2x/(1+x^2)$; domain $\mathbb{R}$.

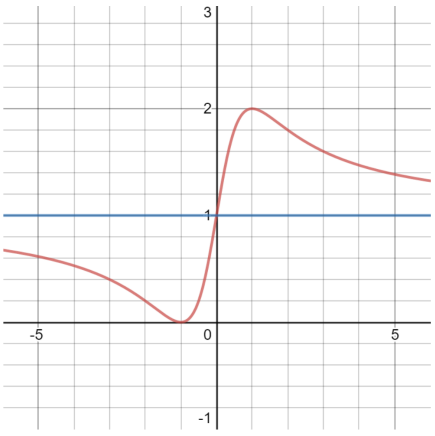
$$f' = \frac{2(1-x^2)}{(1+x^2)^2}, \qquad f'' = \frac{4x(x^2-3)}{(1+x^2)^3}.$$

f' signs $-, +, -$ across $-1, 1$: global min $f(-1) = 0$, max $f(1) = 2$. Direct check: $f \geq 0$, $2 - f = (x-1)^2/(1+x^2) \geq 0$.

f'' signs $-, +, -, +$ across $-\sqrt{3}, 0, \sqrt{3}$. Inflections:

$$(-\sqrt{3}, 1 - \sqrt{3}/2), \quad (0, 1), \quad (\sqrt{3}, 1 + \sqrt{3}/2).$$

$f \to 1^-$ at $-\infty$, 1^+ at $+\infty$; crosses the asymptote at 0.



Lecture 13, p.25.

Sketch order: domain, derivatives, critical points/signs, concavity, asymptotes, key values. Here $f(-x) = 2 - f(x)$: point symmetry about $(0, 1)$.

Three exercises

Jordan. Strict concavity of sin on $[0, \pi/2]$ gives

$$\sin x > 2x/\pi \qquad (0 < x < \pi/2)$$

by comparison with the endpoint chord.

Convex maximum. Strict convexity on $[a, b]$ gives, for $a < x < b$, $f(x) < \alpha f(a) + (1 - \alpha) f(b) \leq \max\{f(a), f(b)\}$. Maximum only at endpoints (possibly both).

Legendre roots. $P_n = (2^n n!)^{-1} D^n[(x^2 - 1)^n]$. Put $Q = (x^2 - 1)^n$.

$$Q^{(k)}(\pm 1) = 0 \quad (0 \leq k < n).$$

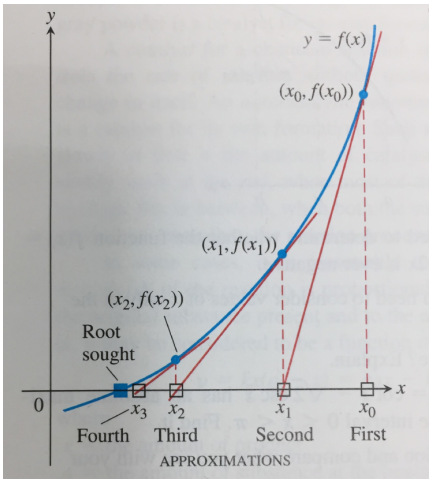
Rolle gives one interior root of Q'. If $Q^{(k)}$ has k interior roots, its endpoint zeros give $k + 1$ gaps, hence $k + 1$ interior roots of $Q^{(k+1)}$. Induct up to n. Since $\deg Q^{(n)} = n$, all n roots are distinct and in $(-1, 1)$.

Newton's method

Linearize at x_n, solve for the tangent's zero:

$$f(x_n) + f'(x_n)(x - x_n) = 0 \implies x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Need $f'(x_n) \neq 0$ at each step.



Lecture 13, p.35.

Ex.1.2.47. $x^2 = 2$:

$$x_{n+1} = \tfrac{1}{2}(x_n + 2/x_n), \qquad x_0 = 1,\ x_1 = 3/2,\ x_2 = 17/12,\ x_3 = 577/408.$$

For $r = \sqrt{2}$, $x_{n+1} - r = (x_n - r)^2/(2x_n)$. After the first step, $x_n \geq r$ and decreases; the positive limit solves $\ell = (\ell + 2/\ell)/2$, so equals r. Error is squared near the root.

Failure examples: e^x gives $x_{n+1} = x_n - 1$ (no root). $\sqrt{|x|}$ gives $x_{n+1} = -x_n$ for $x_n \neq 0$ (oscillation, despite having a root).

Tutorial 3

IVT / MVT

Q7. $F = \cos x - x$: $F(0) > 0 > F(1)$ gives existence in $(0,1)$ by IVT. $F' = -\sin x - 1 < 0$ there; two roots would contradict MVT. Unique.

Q8. Continuous, nonzero f on $[a,b]$:

(a) Opposite signs would give a zero by IVT, so the sign is fixed.
(b) EVT: $m = \min |f| = |f(c)| > 0$, hence $|1/f| \le 1/m$.
(c) Open interval fails: $f(x) = x$ on $(0,1)$.

Q9. MVT: $\sin x - \sin y = \cos c\,(x-y)$, so

$$|\sin x - \sin y| \le |x - y|.$$

Uniform continuity: choose $\delta = \varepsilon$, independent of x, y. More generally, $|f'| \le M$ on an interval gives $|f(x) - f(y)| \le M|x - y|$ (Lipschitz).

Continuity and direct estimates

Q1. $g = x + b$ for $x < 0$, $g = \cos x$ for $x \ge 0$. Continuity at 0: $b = 1$. Then left/right derivatives $1, 0$, so no b makes it differentiable.

Q3. $F = \varphi$ on $\mathbb{Q}$, ψ otherwise; φ, ψ continuous.

$$F \text{ continuous at } a \iff \varphi(a) = \psi(a).$$

Necessity: rational/irrational sequences approaching a force both values to equal $F(a)$. Sufficiency: if common value C, choose a common radius making both pieces within ε of C. For $\sin x, -\sin x$, continuity points $a = k\pi$.

Q10. $(x^3 - 27)/(x^2 - 9) = (x^2 + 3x + 9)/(x + 3)$ off ± 3. Limit at 3 is $9/2$, matching the assignment. At -3, denominator $\to 0$, numerator $\to 9$. Continuity set $\mathbb{R} \setminus \{-3\}$.

Q4(a). If $|f - A|, |g - B| < \varepsilon$, then

$$\max(A,B) - \varepsilon < \max(f,g) < \max(A,B) + \varepsilon,$$
$$\min(A,B) - \varepsilon < \min(f,g) < \min(A,B) + \varepsilon.$$

Use $\delta = \min(\delta_f, \delta_g)$: limits $\max(A,B)$, $\min(A,B)$.

Q4(b). Converse false: $f = 1_{\mathbb{Q}}$, $g = 1 - f$. Max 1, min 0; neither individual limit exists. The pieces exchange roles.

Q2. Rationalize twice, for $0 < |x| < 1$:

$$\left| \frac{\sqrt{4+x} - 2}{x} - \frac{1}{4} \right| = \frac{|x|}{4(\sqrt{4+x} + 2)^2} \le \frac{|x|}{16}.$$

Limit $1/4$; $\delta = \min\{1, 16\varepsilon\}$.

Q11: trigonometric limits

$$\text{(a)}\quad \frac{\sin(\sin 2h)}{h} = \frac{\sin(\sin 2h)}{\sin 2h}\frac{\sin 2h}{2h}\,2 \to 2,$$

$$\text{(b)}\quad \frac{\sin 4y \cot 7y}{y \cot 3y} = \frac{\sin 4y}{y}\frac{\sin 3y}{\sin 7y}\frac{\cos 7y}{\cos 3y} \to \frac{12}{7},$$

$$\text{(c)}\quad \frac{\tan 2x - \sin 2x}{x^3} = \frac{\sin 2x}{x}\frac{2\sin^2 x}{x^2}\frac{1}{\cos 2x} \to 4,$$

$$\text{(d)}\quad \frac{\cos^2 x - \cos^2 a}{x-a} = -\sin(x+a)\frac{\sin(x-a)}{x-a} \to -\sin 2a,$$

$$\text{(e)}\quad \frac{\sqrt{1-\cos 3x^2}}{1-\cos 2x} = \frac{\sqrt{2}|\sin(3x^2/2)|}{2\sin^2 x} \to \frac{3\sqrt{2}}{4}.$$

In (c), factor exactly; first-order approximations cannot recover the cubic term. In (e), keep the absolute value.

Q12: radii / thresholds

(a) $x \to -1^+$: $(x+1)/|x+1| = 1$ exactly. Any $\delta > 0$.

(b) $x \to +\infty$:

$$\left|\frac{7x^2+2}{x^2+1} - 7\right| = \frac{5}{x^2+1} < \frac{5}{x^2}; \quad M = \max\{1, \sqrt{5/\varepsilon}\}.$$

(c) By Q9, for $x > 2$,

$$|\sin\sqrt{x+2} - \sin\sqrt{x-2}| \le \frac{4}{\sqrt{x+2}+\sqrt{x-2}} \le \frac{4}{\sqrt{x-2}}.$$

Limit 0; $M = 2 + 16/\varepsilon^2$.

(d) Put $h = x - a > 0$, $a > 0$. Rationalizing the first numerator term gives

$$E(h) = \frac{\sqrt{h}}{(\sqrt{a+h}+\sqrt{a})\sqrt{2a+h}} + \frac{2}{\sqrt{2a+h}} \to \sqrt{2/a}.$$

For $0 < h < 1$,

$$|E(h) - 2/\sqrt{2a}| \le \frac{\sqrt{h}}{2\sqrt{2}\,a} + \frac{h}{(2a)^{3/2}} \le C\sqrt{h},$$

where $C = (2\sqrt{2}\,a)^{-1} + (2a)^{-3/2}$. Use $\delta = \min\{1, (\varepsilon/C)^2\}$.

Functional relations

Q5(a). Fix $x \ge 0$. Periodicity gives $f(x) = f(x+n) \to L$, hence $f(x) = L$.

Q5(b). Counterexample $f(x) = \sin\sqrt{x}$ on $[0, \infty)$: bounded, continuous, and

$$|f(x+1) - f(x)| \le \frac{1}{\sqrt{x+1}+\sqrt{x}} \to 0.$$

At $(\pi/2 + 2n\pi)^2$ values are 1; at $(3\pi/2 + 2n\pi)^2$, -1. No limit.

Q6: halving and telescoping. Assume $f(x) \to 0$ and $[f(2x) - f(x)]/x \to 0$ as $x \to 0^+$. Given $\varepsilon > 0$, choose $0 < \delta < 1$ with

$$|f(2t) - f(t)| < \frac{\varepsilon}{2} t \qquad (0 < t < \delta).$$

For $0 < x < \delta$, apply at $t = x/2^k$:

$$\begin{aligned} |f(x) - f(x/2^N)| &\le \sum_{k=1}^{N} |f(x/2^{k-1}) - f(x/2^k)| \\ &\le \frac{\varepsilon x}{2} \sum_{k=1}^{N} 2^{-k} < \frac{\varepsilon x}{2}. \end{aligned}$$

Fix x, let $N \to \infty$: $|f(x)| \le \varepsilon x/2 < \varepsilon x$. Thus $f(x)/x \to 0$. Need the first hypothesis to remove the last term (nonzero constants would fail).

Q13. For $0 \le x < 1$, $f(x) = f(x^{2^n}) \to f(0)$ by continuity at 0. At 1, no restriction: $f = 0$ on $[0, 1)$, $f(1) = 1$ is a counterexample. Continuity at 1 forces $f(1) = f(0)$.

Q14. $g(x) = f(x^2)$:

$$\lim_{x \to 0} g(x) = L \iff \lim_{t \to 0^+} f(t) = L.$$

Right to left: from radius η for f, take $\delta = \sqrt{\eta}$. Left to right: from radius δ for g, take $0 < t < \delta^2$, $x = \sqrt{t}$.

Continuity of g need not give continuity of f: $f = 0$ for $t \ge 0$, 1 for $t < 0$ makes $g \equiv 0$.

Q15. Additivity gives $f(0) = 0$ and $f(a + h) - f(a) = f(h)$. Continuity at 0 transports the same ε-δ estimate to every a. In fact the radius is independent of a.

Lecture 14

Antiderivatives and the constant

Definition 1.2.13. F is an antiderivative of f on an interval I if $F' = f$ throughout I. Always specify the interval.

Theorem 1.2.14. If F is one antiderivative on I, all others are $F + C$.

Proof: $(G - F)' = 0$ on the interval; MVT makes $G - F$ constant. On separate components the constants are independent. E.g.

$$\int x\,dx = \begin{cases} x^2/2 + C_-, & -2 < x < -1, \\ x^2/2 + C_+, & 1 < x < 2. \end{cases}$$

Definition 1.2.15. $\int f(x)\,dx$ is the family $\{F + C : C \in \mathbb{R}\}$ on the chosen interval, usually written $F(x) + C$. Not a single function until C is fixed.

Basic formulas ($k \neq 0$; work on intervals where the expressions are defined):

f	$\int f\,dx$	f	$\int f\,dx$
x^p $(p \neq -1)$	$\dfrac{x^{p+1}}{p+1} + C$	$\sin kx$	$-\dfrac{\cos kx}{k} + C$
$\cos kx$	$\dfrac{\sin kx}{k} + C$	$\sec^2 kx$	$\dfrac{\tan kx}{k} + C$
$\csc^2 kx$	$-\dfrac{\cot kx}{k} + C$	$\sec kx \tan kx$	$\dfrac{\sec kx}{k} + C$
$\csc kx \cot kx$	$-\dfrac{\csc kx}{k} + C$		

For arbitrary real p, the power formula is used on $x > 0$; extend only where the real powers make sense.

Ex.1.2.48-50. Integrate finite sums term by term. Useful patterns:

$$\int x^{-2}\,dx = -x^{-1} + C \quad (x \neq 0;\ C_-, C_+),$$

$$\int x^{1/3}\,dx = \tfrac{3}{4}x^{4/3} + C \quad (\mathbb{R}),$$

$$\int \sum_{k=0}^{n} a_k (x + b_k)^k\,dx = \sum_{k=0}^{n} \frac{a_k (x + b_k)^{k+1}}{k+1} + C,$$

$$\int \sum_{k=0}^{n} a_k x (x^2 + b_k)^k\,dx = \sum_{k=0}^{n} \frac{a_k (x^2 + b_k)^{k+1}}{2(k+1)} + C,$$

$$\int \sum_{k=1}^{n} [a_k \cos kx + b_k \sin kx]\,dx = \sum_{k=1}^{n} \frac{a_k \sin kx - b_k \cos kx}{k} + C.$$

Also $\int \sqrt[k]{x + a_k}\,dx = \frac{k}{k+1}(x + a_k)^{(k+1)/k} + C$ on its real intervals, and $\int (x + b_k)^{-k} dx = (x + b_k)^{1-k}/(1 - k) + C$ for $k \geq 2$. Check by differentiating.

Joining the pieces

Ex.1.2.51. A compact answer:

$$\int |x-a|\,dx = \tfrac{1}{2}(x-a)|x-a| + C \qquad (x \in \mathbb{R}).$$

At a, the quotient is $|h|/2 \to 0$. Splitting into two formulas does not give two free constants if the antiderivative is required on all of $\mathbb{R}$.

Ex.1.2.52(a). For $f(x) = \min\{1, |x|\}$,

$$F(x) = C + \begin{cases} x + 1/2, & x \le -1, \\ x|x|/2, & -1 \le x \le 1, \\ x - 1/2, & x \ge 1. \end{cases}$$

Values match at ± 1; derivative there is 1. At 0, derivative 0. Hence $F' = f$ on $\mathbb{R}$.

Ex.1.2.52(b,c): a jump. For either step function $1_{[0,\infty)}$ or $1_{(0,\infty)}$, no antiderivative exists on an open interval about 0: continuity of a proposed primitive joins a constant on the left to $x + C$ on the right, giving one-sided slopes $0, 1$.

On $\mathbb{R} \setminus \{0\}$, however, the primitives are C_- for $x < 0$ and $x + C_+$ for $x > 0$, with independent constants. Removing the joining point changes the problem.

Ex.1.2.53. For $k \in \mathbb{Z}$ and $k\pi \le x \le (k+1)\pi$,

$$\int |\sin x|\,dx = C + 2k + 1 - (-1)^k \cos x.$$

Both adjacent formulas give $C + 2k$ at $x = k\pi$; both derivatives there are 0. Thus this is one primitive on $\mathbb{R}$. Its value increases by 2 each half-period; the primitive is not periodic.

Lecture 15

Sums, partitions, tags

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2.$$

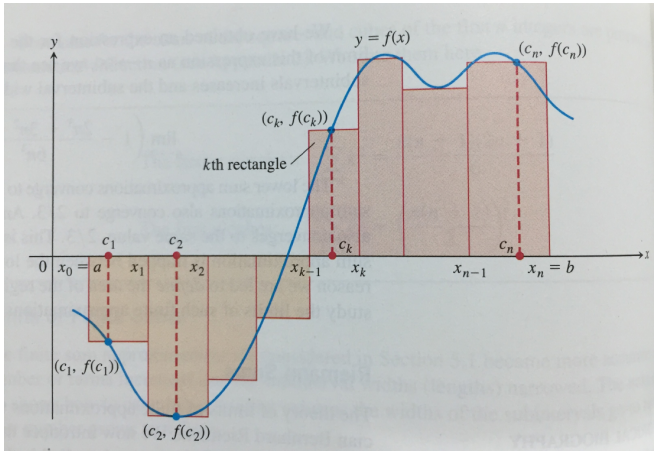
Two standard finite sums: $\sum_{k=1}^{n} 2^{-k} = 1 - 2^{-n}$ and $\sum_{k=1}^{n} [k(k+1)]^{-1} = 1 - (n+1)^{-1}$.

Definition. A partition P of $[a, b]$:

$$a = x_0 < x_1 < \cdots < x_n = b, \quad \Delta x_k = x_k - x_{k-1}, \quad \|P\| = \max_k \Delta x_k.$$

Choose tags $c_k \in [x_{k-1}, x_k]$. The Riemann sum is

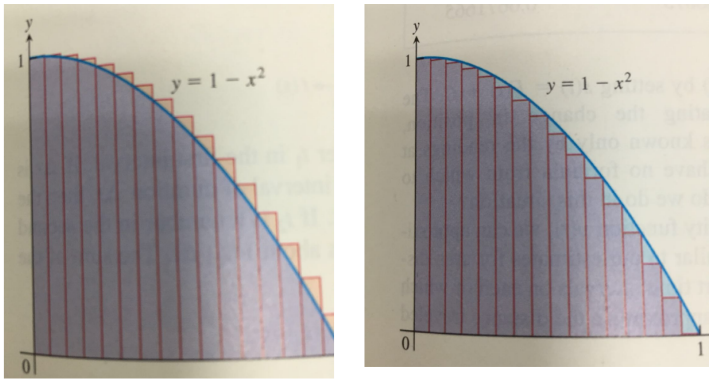
$$S(f, P, \mathbf{c}) = \sum_{k=1}^{n} f(c_k)\Delta x_k.$$



Lecture 15, p.12. Negative heights contribute negative terms.

Equal widths $\Delta = (b-a)/n$: left, midpoint and right tags are $a+(k-1)\Delta$, $a+(k-1/2)\Delta$, $a+k\Delta$. For bounded f, let $m_k = \inf_{[x_{k-1},x_k]} f$, $M_k = \sup_{[x_{k-1},x_k]} f$. Then

$$L(f,P) = \sum m_k \Delta x_k \le S(f,P,\mathbf{c}) \le \sum M_k \Delta x_k = U(f,P).$$



Lecture 15, pp.4-5. Upper / lower rectangles for $1-x^2$ on $[0,1]$.

The limit must ignore the choices

Definition (Riemann integral). $f : [a,b] \to \mathbb{R}$ is integrable with integral J if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall P \ \forall \mathbf{c} : \qquad \|P\| < \delta \implies |S(f,P,\mathbf{c}) - J| < \varepsilon.$$

Write $J = \int_a^b f(x)\,dx$. This is a number; $\int f(x)\,dx$ is a family of functions.

- Need **all** sufficiently fine partitions and **all** tags. One convergent sequence of sums is not enough to prove integrability.
- $n \to \infty$ does not imply $\|P\| \to 0$: keep $[0,1/2]$ as one cell and subdivide $[1/2,1]$ more and more.
- Finite endpoint changes do not affect J: the change in a sum is at most $(|\Delta f(a)| + |\Delta f(b)|)\|P\|$.

Ex.2.1.1. For x^2 on $[0,1]$, right sums give

$$\frac{1}{n^3}\sum_{k=1}^{n} k^2 = \frac{1}{6}(1+1/n)(2+1/n) \to \frac{1}{3}.$$

To check arbitrary partitions/tags, use $F = x^3/3$. MVT on each cell gives $\Delta F = \xi_k^2 \Delta x_k$. Therefore

$$\left| S - \frac{1}{3} \right| \le \sum |c_k^2 - \xi_k^2| \Delta x_k \le 2\|P\| \sum \Delta x_k = 2\|P\|.$$

So $\delta = \varepsilon/2$ works. Likewise the right sums for $1 - x^2$ tend to $2/3$:

$$\frac{1}{n} \sum_{k=1}^{n} [1 - (k/n)^2] = 1 - \frac{1}{6}(1 + 1/n)(2 + 1/n) \to \frac{2}{3}.$$

Tutorial (Homework 3)

This is the new five-page sheet, not the earlier Tutorial 3.

Q1: limits

(a) $s_n = \sum_{k=1}^{n} (1/k - 1/(k+1)) = 1 - 1/(n+1) \to 1$. Given $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$; $n > N \Rightarrow |s_n - 1| < \varepsilon$.

(b) Put $a = 1 + t$, $t > 0$. For $n \ge 2$,

$$a^n \ge \binom{n}{2} t^2 \implies 0 \le \frac{n}{a^n} \le \frac{2}{(n-1)t^2} \to 0.$$

Choose an integer $N > \max\{2, 1 + 2/(\varepsilon t^2)\}$.

(c) Cancel before taking the limit:

$$\frac{1}{1-x} - \frac{3}{1-x^3} = -\frac{x+2}{x^2+x+1} \to -1.$$

For $0 < |x-1| < 1/2$, error $= |x-1||x+1|/(x^2+x+1) \le (5/2)|x-1|$. Use $\delta = \min\{1/2, 2\varepsilon/5\}$.

(d) Let $s = \sqrt{1+x} + \sqrt{1-x}$. Rationalize twice:

$$\frac{s-2}{x^2} = -\frac{2}{(\sqrt{1-x^2}+1)(s+2)} \to -\frac{1}{4}.$$

(e) $\cos 2x - 4\cos x + 3 = 2(1 - \cos x)^2 = 8\sin^4(x/2)$; dividing by x^4 gives limit $1/2$.

Q2-Q4: auxiliary functions and asymptotes

Q2(a). $f \in C[0,1]$, differentiable inside, $f(0) = 0$, $f(1) = 1$. IVT gives $t \in (0,1)$ with $f(t) = 1/2$. MVT on $[0,t]$, $[t,1]$ gives $0 < \xi < t < \eta < 1$ and

$$f'(\xi) = \frac{1}{2t} > 0, \quad f'(\eta) = \frac{1}{2(1-t)} > 0; \qquad \frac{1}{f'(\xi)} + \frac{1}{f'(\eta)} = 2.$$

Q2(b). Set $H(x) = (1-x)f(x)$. Since $H(0) = H(1) = 0$, Rolle gives $(1-c)f'(c) - f(c) = 0$. The product is chosen to produce the requested derivative.

Q3(a). $f = \sqrt{x^2+2x} - x$, domain $(-\infty, -2] \cup [0, \infty)$:

$$x \to +\infty: \quad f = \frac{2}{\sqrt{1+2/x}+1} \to 1,$$
$$x \to -\infty: \quad f - (-2x-1) = \sqrt{x^2+2x} + x + 1 \to 0.$$

For the second line, $\sqrt{x^2+2x} + x = 2x/(\sqrt{x^2+2x} - x) \to -1$. Horizontal $y = 1$ on the right; oblique $y = -2x - 1$ on the left. Endpoint values finite, so no vertical asymptote. Use $\sqrt{x^2} = |x|$.

Q3(b). $g = \sqrt{x^2+1}/x$: limits $1, -1$ at $+\infty, -\infty$; limits $+\infty, -\infty$ at $0^+, 0^-$. Horizontal lines $y = 1, -1$, vertical $x = 0$; no oblique lines.

Q3(c).

$$h = \frac{(x-1)(x^2+1)}{(x-1)(x-2)} = x + 2 + \frac{5}{x-2} \quad (x \neq 1, 2).$$

Vertical $x = 2$, oblique $y = x + 2$ at both infinities. $x = 1$ is a removable hole (limit -2), not a vertical asymptote. No horizontal line.

Q4. As in Tutorial 3, Q8: IVT fixes the sign; EVT gives $m = \min |f| > 0$, hence $|1/f| \leq 1/m$. On $(0, 1)$, $f(x) = x$ shows the open-interval conclusion fails.

Q5-Q8: derivatives and extrema

Q5. $F = ax + b$ for $x < 0$, $F = \cos x$ for $x \geq 0$.

(a) Continuity: $b = 1$, any a.
(b) One-sided derivatives $a, 0$; differentiable iff $(a, b) = (0, 1)$, then $F'(0) = 0$.
(c) For these values, $F' = 0$ on the left, $-\sin x$ on the right; continuous at 0. But the one-sided difference quotients of F' tend to $0, -1$, so $F''(0)$ does not exist.

Q6(a). For $y = \sin(x^2)/\sqrt{1+x^2}$,

$$y' = \frac{2x\cos(x^2)}{\sqrt{1+x^2}} - \frac{x\sin(x^2)}{(1+x^2)^{3/2}}.$$

Q6(b). $y = \arcsin[2x/(1+x^2)]$:

$$\sqrt{1 - \left(\frac{2x}{1+x^2}\right)^2} = \frac{|1-x^2|}{1+x^2}, \qquad y' = \begin{cases} 2/(1+x^2), & |x| < 1, \\ -2/(1+x^2), & |x| > 1. \end{cases}$$

Not differentiable at ± 1: left/right slopes are $-1, 1$ at -1 and $1, -1$ at 1. The square root needs the absolute value.

Q6(c). Leibniz, $P = (1+x^2)\cos x$, $n \geq 2$:

$$P^{(n)} = (1+x^2)\cos(x + n\pi/2) + 2nx\cos(x + (n-1)\pi/2) + n(n-1)\cos(x + (n-2)\pi/2).$$

Q7. $x^2 + xy + y^2 = 3$, $y(1) = 1$:

$$2x + y + (x + 2y)y' = 0 \implies y'(1) = -1,$$
$$2 + 2y' + 2(y')^2 + (x + 2y)y'' = 0 \implies y''(1) = -2/3.$$

Tangent $y - 1 = -(x - 1)$; normal $y - 1 = x - 1$. Linearization $L(x) = 2 - x$, so $y(1.02) \approx 0.98$.

Q8. $f = x^3|x - 1|$ on $[-1, 2]$:

$$f' = \begin{cases} x^2(3 - 4x), & x < 1, \\ x^2(4x - 3), & x > 1. \end{cases}$$

x	-1	0	$3/4$	1	2
$f(x)$	-2	0	$27/256$	0	8

Interior critical points $0, 3/4, 1$. Signs of f': $+$ up to $3/4$ (except zero at 0), then $-$ up to 1, then $+$. Local max at $3/4$, local min at 1; 0 is neither. Absolute min -2 at -1, max 8 at 2.

Q9-Q11: what derivative bounds force

Q9(a). Differentiable f on an open interval I:

$$|f'| \le M \iff |f(x) - f(y)| \le M|x - y| \quad (x, y \in I).$$

Forward: MVT. Reverse: divide by $|h|$, then let $h \to 0$.

Q9(b). If $f(b) - f(a) = M(b - a)$, for $a \le x \le b$ the Lipschitz bounds give

$$f(b) - M(b - x) \le f(x) \le f(a) + M(x - a).$$

The two bounds are equal. Thus $f(x) = f(a) + M(x - a)$ throughout $[a, b]$.

Q9(c). From $|g(x) - g(y)| \le C|x - y|^2$,

$$\left|\frac{g(x + h) - g(x)}{h}\right| \le C|h| \to 0.$$

Hence $g' = 0$ and g is constant by MVT. With exponent 1, $g(x) = x$ is a counterexample.

Q10. Zeros $x_n \ne c$, $x_n \to c$; f differentiable at c.

(a) Continuity gives $f(c) = 0$; along x_n, $[f(x_n) - f(c)]/(x_n - c) = 0$, so $f'(c) = 0$.
(b) If f is differentiable near c and $f''(c)$ exists, Rolle between x_n, c gives $z_n \ne c$ with $f'(z_n) = 0$ and $z_n \to c$. Therefore $f''(c) = \lim[f'(z_n) - f'(c)]/(z_n - c) = 0$.

Q11 (Darboux property). Given $f'(a) < \lambda < f'(b)$, set $H = f - \lambda x$. Then $H'(a) < 0 < H'(b)$, so neither endpoint minimizes H on $[a, b]$. EVT + Fermat gives an interior minimum c with $H'(c) = 0$, i.e. $f'(c) = \lambda$.

No continuity of f' used. A derivative may oscillate, but cannot have a jump.

Supplementary exercises

S1: interpolation error. $f \in C[a, b]$, twice differentiable inside, $|f''| \le M$. Let L be the chord interpolant and set

$$E = f - L, \qquad q(x) = \frac{M}{2}(x - a)(b - x).$$

Both E and q vanish at the endpoints. Since

$$(E - q)'' = f'' + M \ge 0, \qquad (E + q)'' = f'' - M \le 0,$$

the chord inequalities give $E - q \le 0$ and $E + q \ge 0$. Thus

$$|f(x) - L(x)| \le \frac{M}{2}(x-a)(b-x) \le \frac{M(b-a)^2}{8}.$$

Smallest constant $K = 1/8$: equality at the midpoint for $f(x) = \frac{M}{2}(x-a)(x-b)$, whose chord is zero.

S2: a primitive vs. an integral.

$$g_\lambda(x) = \begin{cases} 2x\sin(1/x) - \cos(1/x), & x \neq 0, \\ \lambda, & x = 0. \end{cases}$$

(a) Put $F(x) = x^2 \sin(1/x)$ off 0, $F(0) = 0$. We know $F'(0) = 0$; $F' = g_0$ everywhere. If $G' = g_\lambda$, then off 0, $G = F + C_\pm$. Continuity at 0 forces one common constant, hence $G'(0) = 0$. Therefore

$$\lambda = 0 \text{ only;} \qquad \text{all primitives: } F + C.$$

(b) All λ are Riemann integrable on $[-1, 1]$. Prove it for arbitrary tagged sums, using MVT rather than an integration formula.

Set $J = F(1) - F(-1) = 2\sin 1$ and $B = 3 + |\lambda|$, so $|g_\lambda| \le B$ and $|F'| \le 3$ on $[-1, 1]$. Given $\varepsilon > 0$, choose

$$r = \min\left\{\frac{1}{4}, \frac{\varepsilon}{16(B+3)}\right\}, \quad A = 2 + 2/r + 1/r^2, \quad \delta = \min\{r, \varepsilon/(4A)\}.$$

For any $\|P\| < \delta$, MVT gives $\Delta F_k = g_0(\xi_k)\Delta x_k$, so

$$S(g_\lambda, P, \mathbf{c}) - J = \sum_k [g_\lambda(c_k) - g_0(\xi_k)]\Delta x_k.$$

Split cells into two groups:

- Cells meeting $[-r, r]$: total width $\le 2r + 2\|P\| < 4r$. Error $\le 4r(B+3) \le \varepsilon/4$.
- Other cells: c_k, ξ_k lie on the same side, with $|x| \ge r$. Here $|g_0'| \le A$, so their total error is $\le A\|P\| \sum \Delta x_k < 2A\delta \le \varepsilon/2$.

Total error $< \varepsilon$, independently of all tags. Hence

$$\int_{-1}^{1} g_\lambda(x)\,dx = 2\sin 1 \qquad (\lambda \in \mathbb{R}).$$

Changing one value preserves the integral, but can destroy the existence of a primitive.